Pulseq Siemens Workflow

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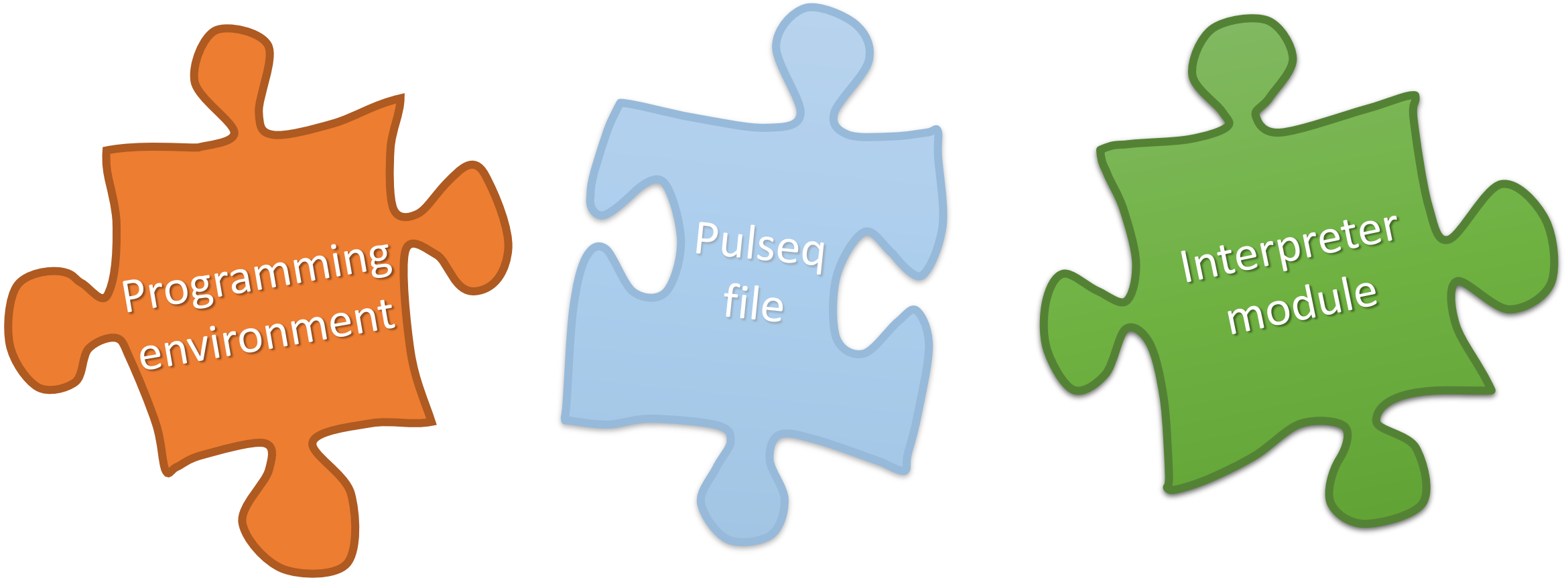
February 26, 2026

Outline

- Features of the ***Pulseq*** environment that are Siemens specific:
  - Accurate peripheral nerve stimulation prediction using the SAFE model
  - Acoustic resonances analysis with gradient system files
  - Relative (!) SAR and B1-rms

- Using ***Pulseq*** on Siemens scanners:
  - ***Pulseq*** Interpreter Sequence
  - IDEA simulation
  - Scanner set up and special use cases
  - Note on shimming

Pulseq : pieces of the puzzle



Pulseq PNS prediction

- Peripheral neural stimulation (PNS) prediction on Siemens is based on the so-called SAFE model

  F.X. Herbank and M. Gebhardt. SAFE-Model - A New Method for Predicting Peripheral Nerve Stimulations in MRI. ISMRM 2000, #2007. https://cds.ismrm.org/ismrm-2000/PDF7/2007.PDF

- Open-source implementation by Filip Szczepankiewicz and Thomas Witzel: https://github.com/filip-szczepankiewicz/safe_pns_prediction

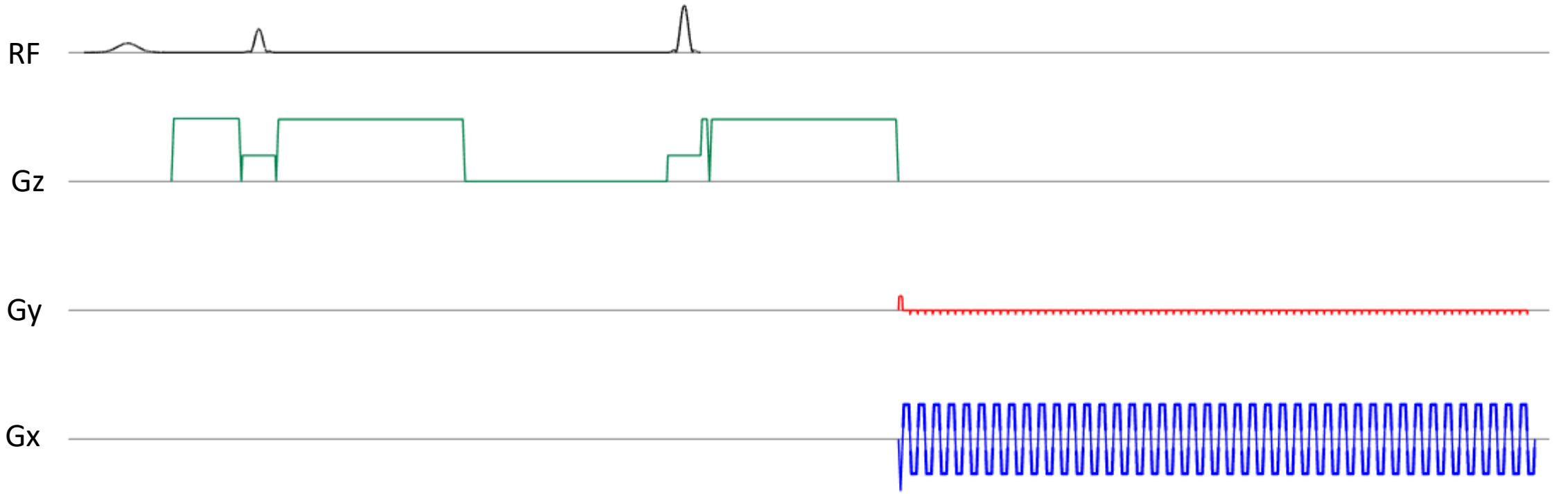
- Direct interface in Matlab-Pulseq

Caveat & Solution

- Need scanner-specific model parameters
- Extract your gradient system description file from IDEA
  - Type “sys” in the IDEA shell selecting your system
    (you may need to select something else first to see the verbose output)
  - Note the strings after “GPA Type” and “GC Type”
  - On Numaris 4
    - Go to C:\MIDEA\N4_VE####\n4\pkg\MrServers\MrMeasSrv\Config\InitMeas then pick the file named MP_GPA_<your_GPA>_<your_coil>.asc and copy it somewhere where your Matlab or Python can read it
  - On Numaris X
    - Go to C C:\MIDEA\NXVA###_###\Config\MriProduct\Measurement\InitMeas then pick the file named MP_GradSys_<your_GPA>_<your_coil>.asc
- Stimulation parameters on the scanner are sometimes changed after updates

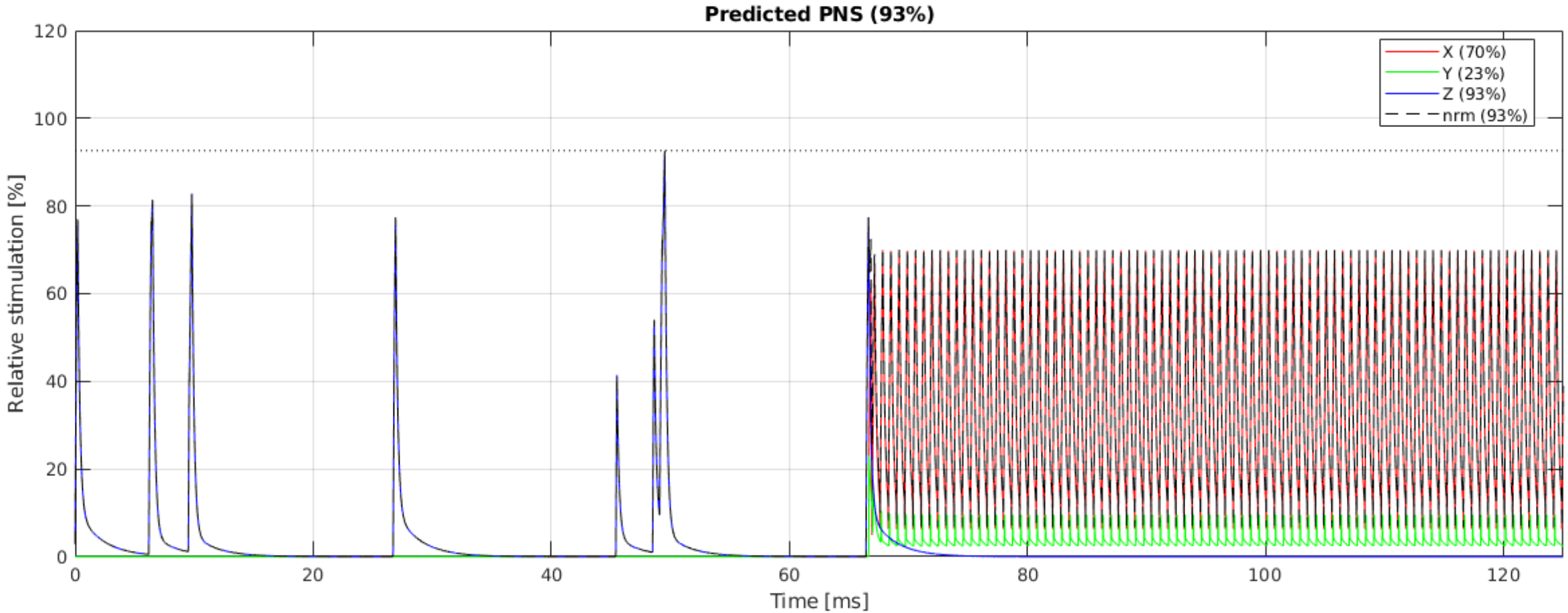
Example: DW-EPI

- bFactor=1000, TE=78ms, Gmax=38mT/m SR=180T/m/s



PNS Stimulation: DW-EPI

seq.calcPNS('idea/asc/MP_GPA_K2309_2250V_951A_AS82.asc');



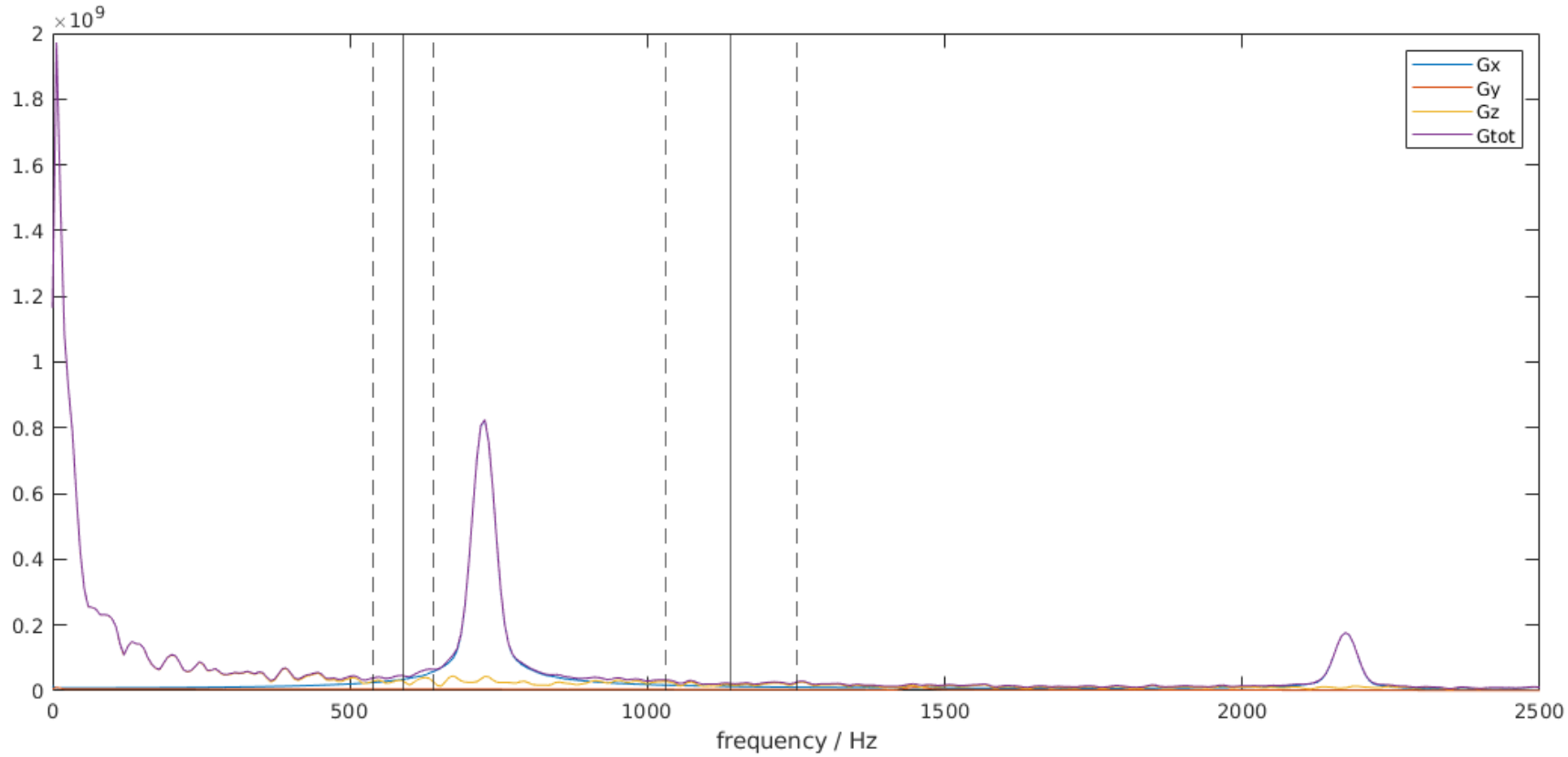
Acoustic resonance analysis

- Extract your gradient system description file from IDEA as above

- Create your sequence in memory
  (populate the “seq” object with events)

- Run seq.gradSpectrum(…)
  - You can provide your gradient system .asc file to see resonances marked

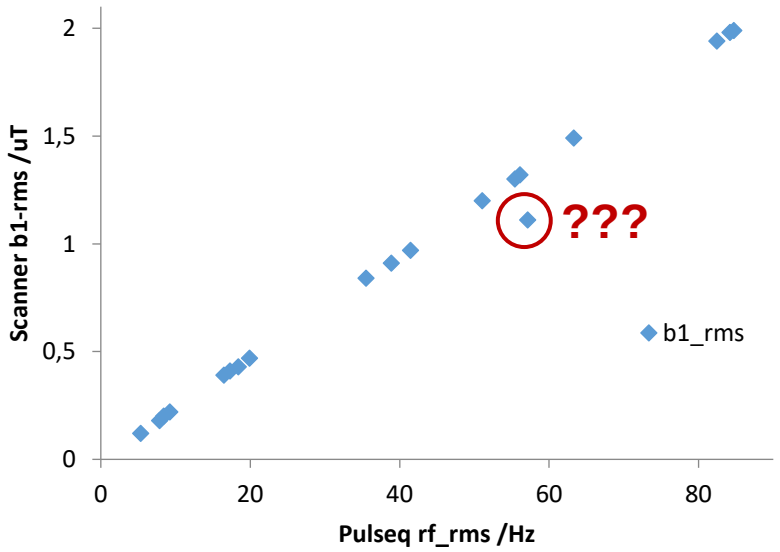
Acoustic analysis: DW-EPI



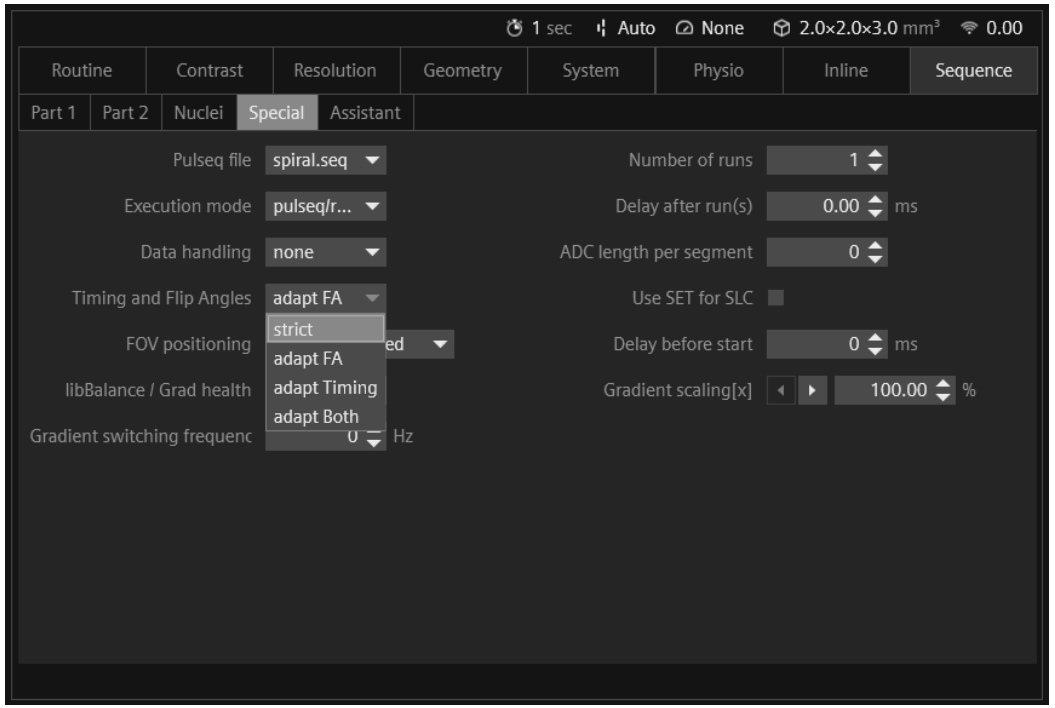
seq.sound(); % another option for acoustic assessment in Matlab ;-)

Calculate B1-RMS and Relative SAR

- Use seq.calcRfPower(…) to calculate relative SAR and rf_rms
- Pulseq prediction is highly accurate
  - Watch out for “Flip Angles -> strict” (not safety- but reproducibility-relevant)



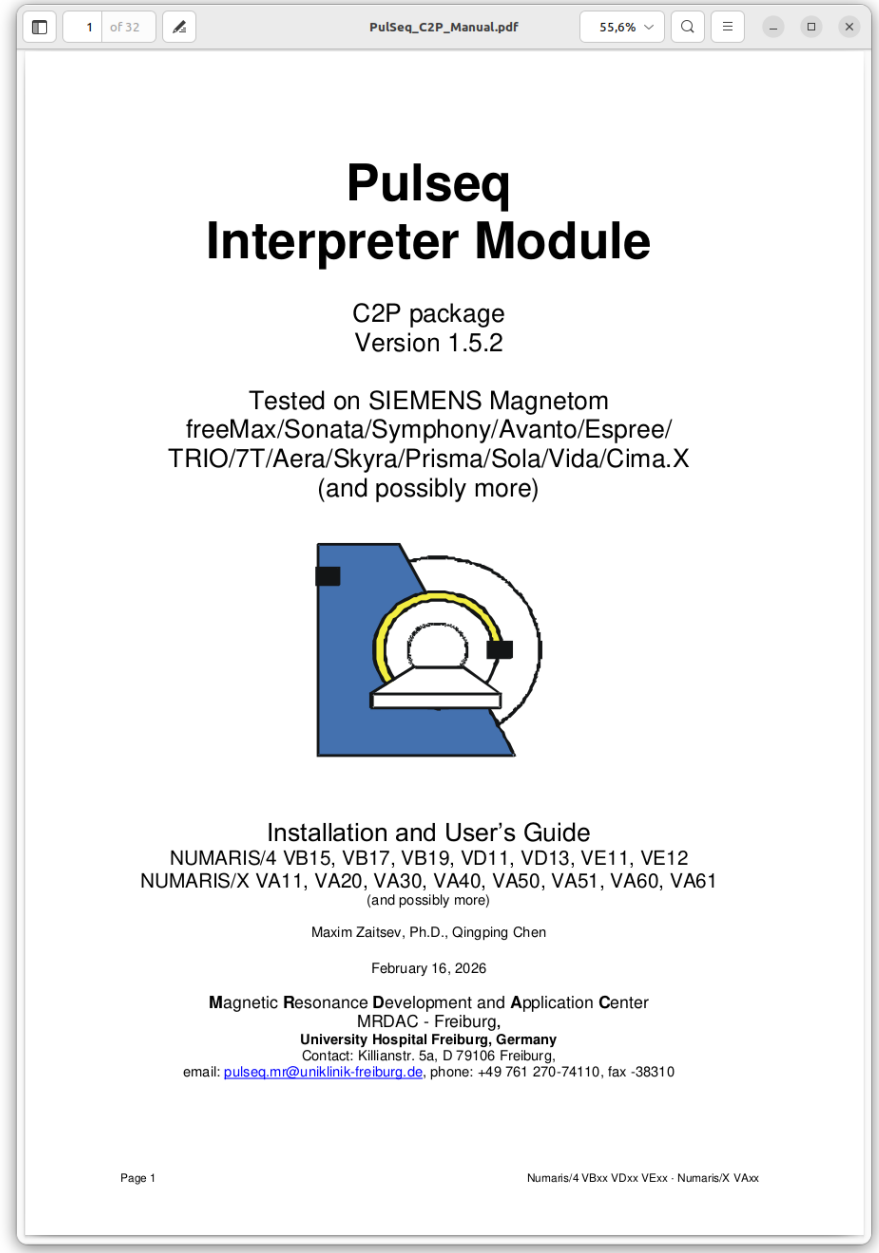
B1 RMS of all Pulseq Demo sequences
courtesy of Cai Wan



Pulseq Siemens interpreter

- Just a “normal” sequence
  - Loads its “content” from a Pulseq file
  - Almost all aspects of the sequence are pre-defined in the Pulseq file
  - FOV positioning and rescaling possible
- Based on miniFlash
  - No product code
  - No hacks, no backdoors
- Distributed as a C2P package in **source form**

- Standard SAR calculation and PNS monitoring
- “Gradient Health” libBalance applicable to all* sequences
- Design-time PNS and acoustic resonance analysis
- **Safety equal or higher than a typical IDEA sequence**



PulSeq_C2P_Manual.pdf

Pulseq
Interpreter Module

C2P package
Version 1.5.2

Tested on SIEMENS Magnetom
freeMax/Sonata/Symphony/Avanto/Espree/
TRIO/7T/Aera/Skyra/Prisma/Sola/Vida/Cima.X
(and possibly more)

Installation and User’s Guide
NUMARIS/4 VB15, VB17, VB19, VD11, VD13, VE11, VE12
NUMARIS/X VA11, VA20, VA30, VA40, VA50, VA51, VA60, VA61
(and possibly more)

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February 16, 2026

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Page 1 Numaris/4 VBxx VDxx VExx · Numaris/X VAxx

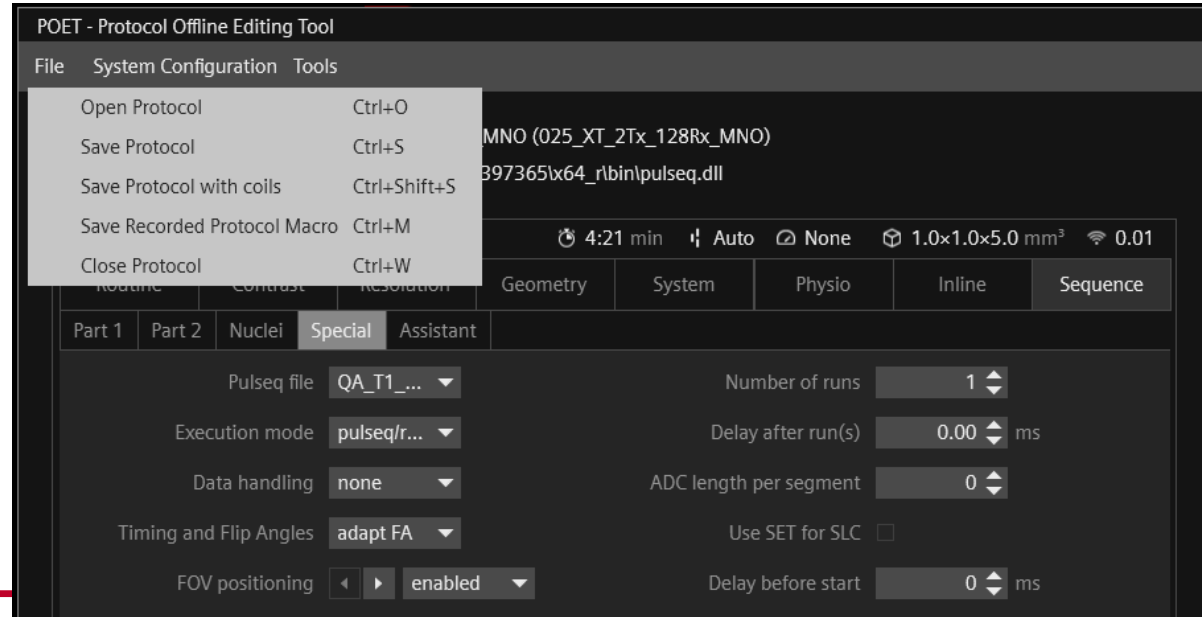
* longer sequences are known to crash libBalance

Pulseq on Siemens platforms

- Over 200 C2P sites
- *Pulseq* works well on:
  - All Numaris4 platforms (tested on vb17…ve12u) and numerous hardware platforms (Symphony, Trio, 7T, 3T Connectom, Skyra, Prisma,…)
  - Most released NumarisX versions (xa11, xa20, xa30, xa40, xa5x, xa6x)
    - No official support for xb10 or xa80 yet (and maybe others)

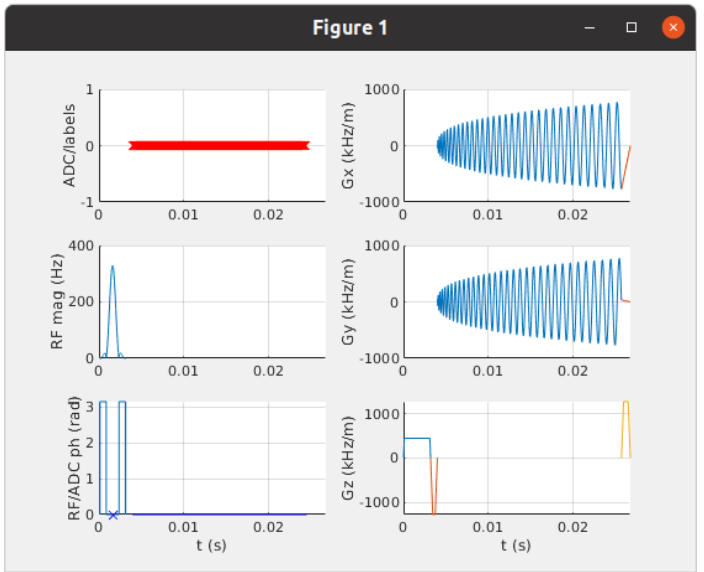
IDEA Simulation

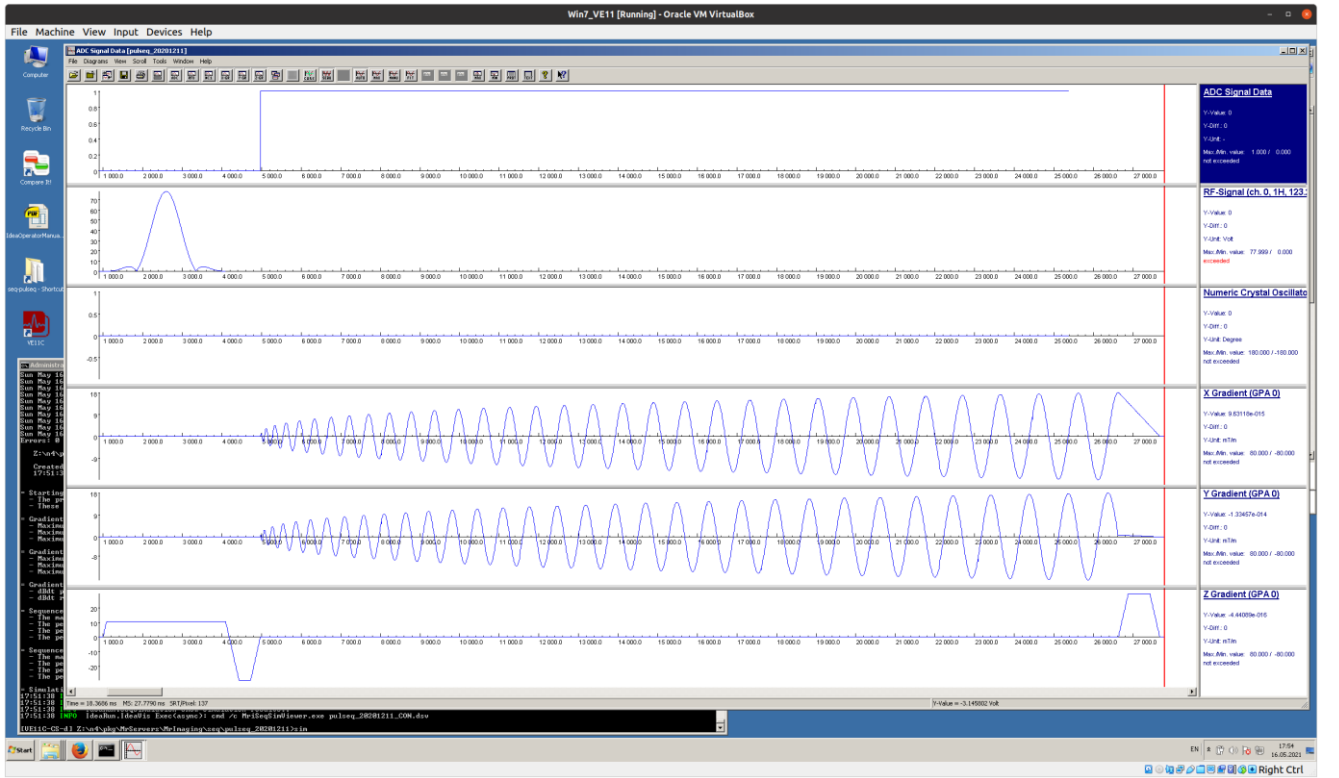
- Compile Windows (debug) version of the interpreter in IDEA
- Create a directory C:\ProgramData\Siemens\Numaris\MriCustomer\CustomerSeq\pulseq and populate it with sequences
- Open POET, select the desired sequence and simulate it
- To run a simulation from console save the protocol **with coils** (!) in POET and then start sim my_pulseq_protocol.pro



IDEA simulation #2

Accelerate debugging cycles by mapping your Matlab/Python sequence directory to your IDEA-VM (see C2P manual)



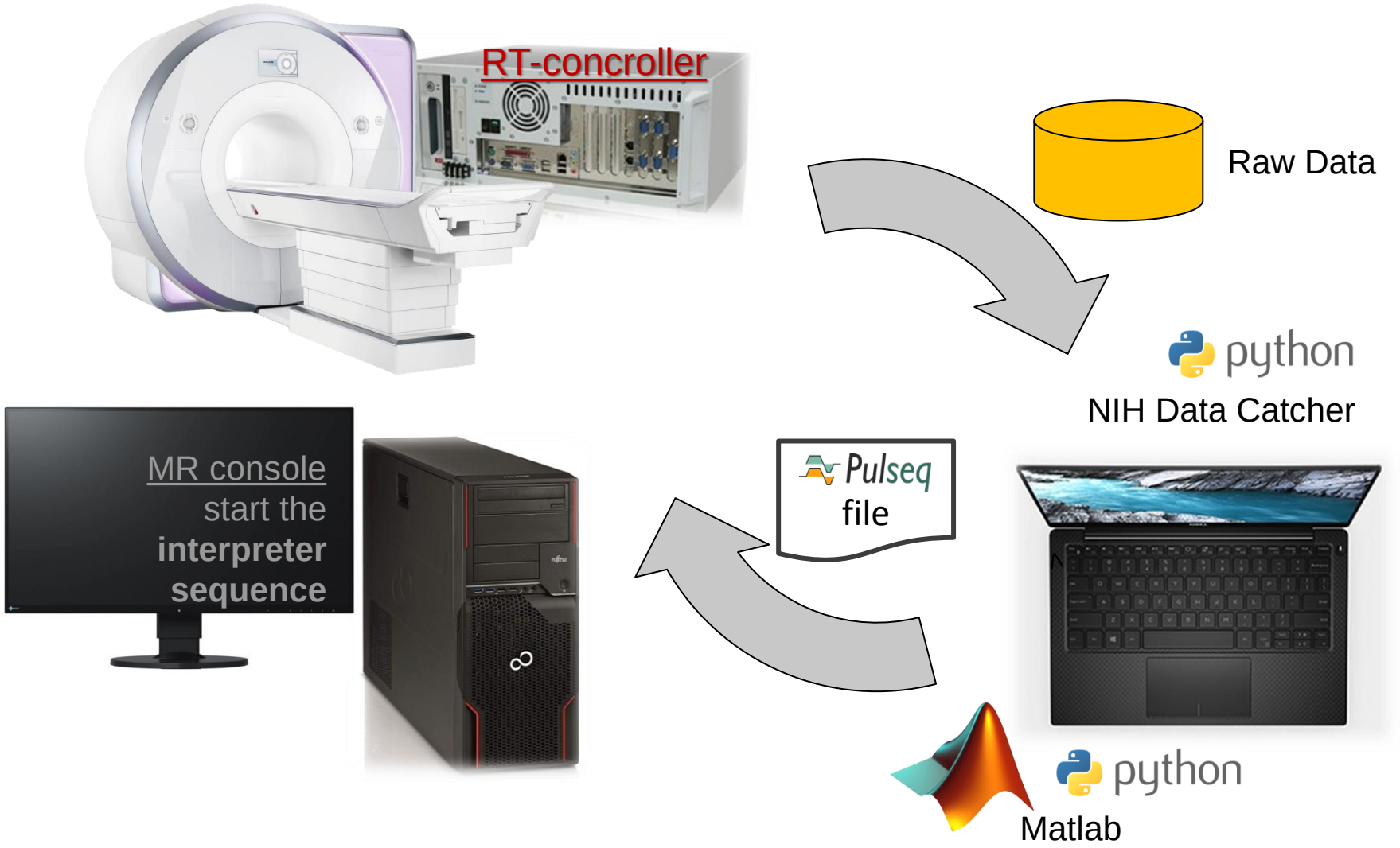


Debugging if simulation fails:

- Watch IDEA console or LogViewer for errors and warnings
- To increase verbosity change const MessageType MSG_LEVEL =… in ExternalSequence.h

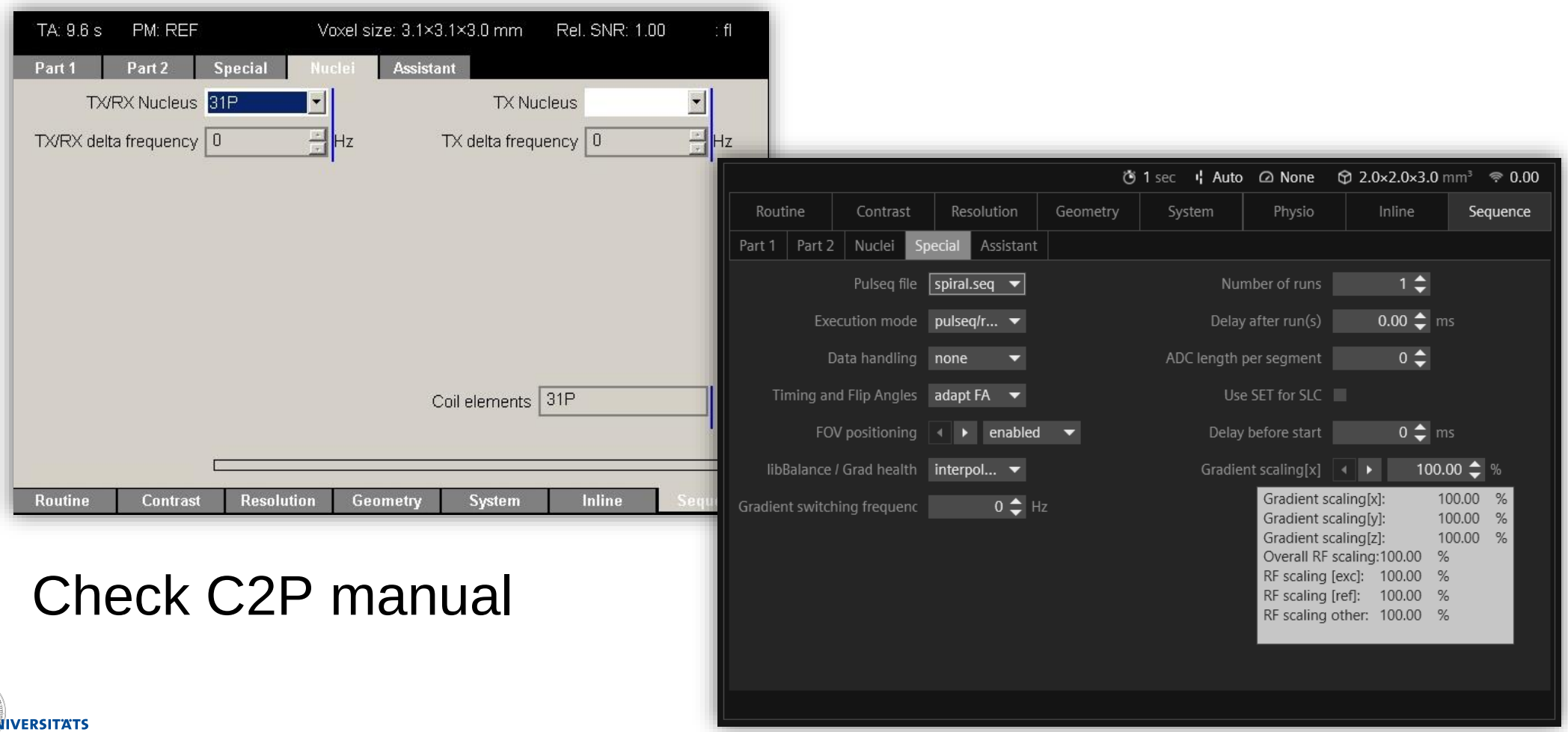
Pulseq on Siemens scanners

- *Optional initial step: connect your PC to the scanner*
- Save the .seq file(s) on the scanner
- Run the interpreter_sequence on the scanner
- *Optional step: stream raw data to your PC with NIH_DataCatcher*
- or export raw data manually



Pulseq interpreter parameters

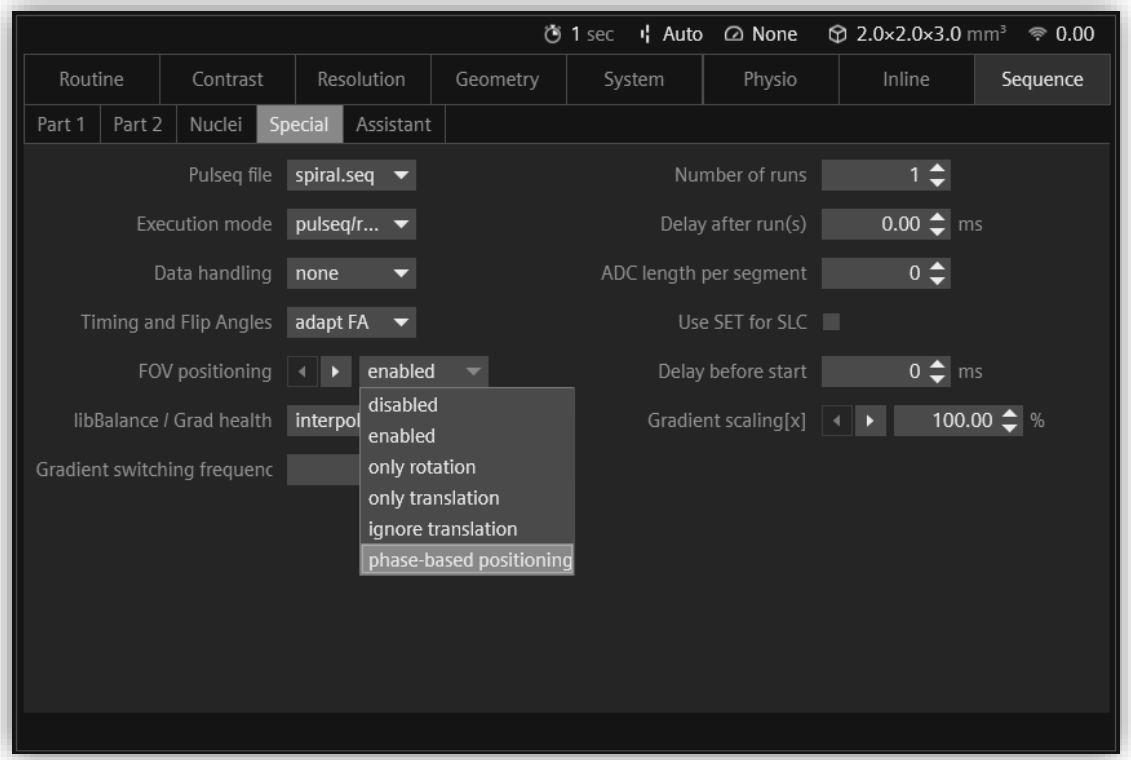
- Yes, you can still change few things on the console



- Check C2P manual

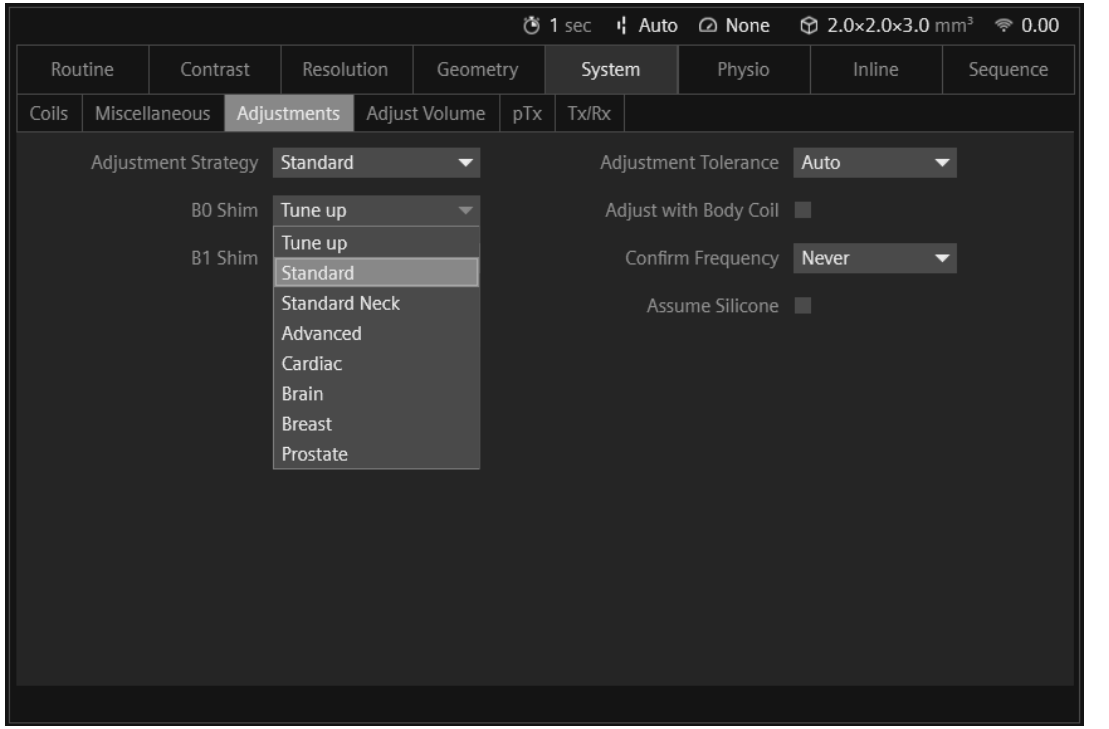
Update on FOV Positioning

- X-Nuclei / MNS: there is a bug in the underlying libraries preventing the conventional Pulseq FOV positioning from working for MNS (all Numaris4 VExx and all NumarisX prior to VBxx)
- New <u>experimental</u> option in v152: phase-based FOV positioning
  - Only works for constant gradients during RF and ADC
  - Should resolve the MNS bug
  - In future: better FOV positioning for non-Cartesian imaging



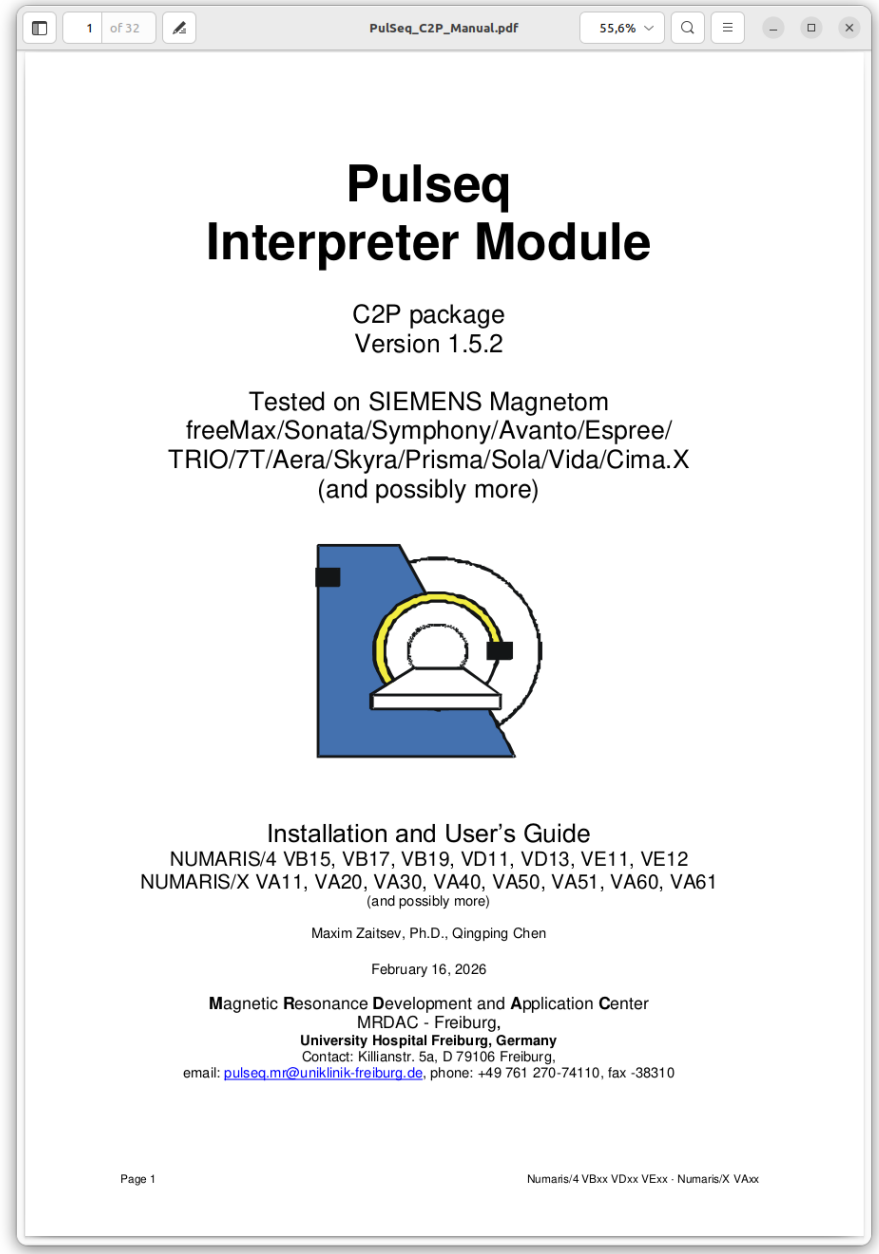
Note on Shimming

- Many people report problems with shimming
  1. Make sure your **FOV is correctly described** in Definitions
  2. Select “Standard” under System->Adjustments->B0 Shim (Pulseq defaults to “Tune up”)
  3. *Copy the protocol if needed*
- Any other shim mode may also be used



Conclusions & Outlook

- Pulseq Interpreter is full of features
  - Read the C2P manual: 32 pages of infos
- Numerous minor improvements in v152
  - Automatic selection on the previous directory in the sequence selector
- Open Sequence (OS) interpreter
  - Proof-of-concept status
  - Multiple sequences tested
  - Takes 1-2 years of development to catch up
  - Stop by at Mojtaba Shafiekhani's poster at the ISMRM 2026 in Cape Town



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Acknowledgements:

Berkin Bilgic
Borjan Gagoski
Douglas Noll
Frank Zijlstra
Imam Shaik
Jeff Fessler
Jon-Fredrik Nielsen
Juergen Hennig
Mojtaba Shafiekhani
Moritz Zaiss
Naveen Murthy
Niklas Wehkamp
Qiang Liu
Qingping Chen
Scott Peltier
Sebastian Littin
Will Grissom
Yogesh Rathi

THANK YOU FOR YOUR ATTENTION!

Pulseq MATLAB

pypulseq Python

Supported by NIH U24-NS120056 (Nielsen, Zaitsev)
and R01-EB032378 (Rathi, Bilgic)

NIH National Institutes of Health